



BOILEAU GUILLAUME

Ingenieur Validation Connectivité & Systèmes | Tests, Logs, Defects, Python, Linux, Hardware/Software

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EXPERIENCE

Software Engineer – System Integration, Validation & Production Follow-up

VEV Platform Services France

📅 2025 – 2026

📍 Valbonne, France

Context: Industrial software and connected infrastructure environment for an EV charging CPMS platform linked to operational charging stations, field equipment and hardware/software interfaces.

Objective: Validate service behaviour, analyse anomalies and support reliable operation of connected systems through tests, checks, logs, operational follow-up and technical coordination.

- **Connected systems validation:** Verified service behaviour, operational continuity and data-flow consistency between platform services and field infrastructure.
- **Connectivity-related issue analysis:** Investigated communication failures, interrupted exchanges, inconsistent data flows and anomalies affecting connected equipment workflows.
- **Logs, traces & defects:** Analysed operational symptoms, service behaviour, issue reports and technical traces to identify defects and support resolution.
- **Hardware/software collaboration:** Worked on software services interacting with charging-station infrastructure, field equipment and operational constraints.
- **Manual & operational tests:** Performed validation checks in production-like or live operational conditions to assess service continuity and expected behaviour.
- **Automation & scripts:** Developed and maintained TypeScript, Angular and Node.js components supporting operational workflows; strong Python/Linux background for test automation.
- **Data exchange validation:** Implemented, monitored and checked FTP-based operational data exchange services.
- **Quality follow-up:** Documented findings, clarified incidents, tracked corrective actions and supported technical reporting with stakeholders.
- **Agile delivery:** Used Zenhub/GitHub-style workflows for backlog follow-up, task tracking, issue prioritisation and iterative delivery.

Senior R&D Engineer – Validation, Test Strategy & Automation

CNES, Laboratoire Lagrange, Observatoire de la Côte d'Azur

📅 2023 – 2025

📍 Nice, France

Context: Engineering activities for the LISA ESA/NASA space mission, involving complex software chains, data-processing workflows, simulation, model integration and validation campaigns.

Objective: Define validation strategies, execute test-like analysis campaigns, automate checks and analyse results under strong consistency, traceability and robustness constraints.

- **Validation strategy:** Defined validation logic, comparison campaigns, consistency checks and acceptance criteria for complex technical outputs.
- **Test campaign execution:** Prepared, executed and analysed validation campaigns involving large datasets, simulated outputs and complex software behaviours.
- **Python automation:** Developed CATpyLISA, a Python platform for large-scale model integration, comparison, validation and technical review. [GitLab: CATpyLISA](#)
- **Anomaly identification:** Detected deviations, inconsistencies, limiting factors and technical risks affecting validation quality and result reliability.
- **Traceability:** Linked requirements, assumptions, inputs, processing steps and outputs through documented and versioned workflows.
- **Linux/Git workflows:** Used Linux, Python, Git/GitLab and reproducible environments to automate analyses and support repeatable validation.
- **Reporting quality:** Produced validation summaries, technical notes, analysis reports and review material for project stakeholders.
- **Cross-functional work:** Collaborated with international contributors across software, systems, data, modelling and instrumentation constraints.
- **Continuous improvement:** Improved validation workflows, documentation structure, analysis tools and review practices.

Data Scientist – Trace Analysis, Debugging Logic & Performance Validation

Universiteit Antwerpen, Belgium

📅 2021 – 2023

📍 Antwerp, Belgium

Context: Performance analysis for precision instrumentation in an international technical environment linked to detector sensitivity and system diagnostics.

Objective: Identify, characterize and mitigate abnormal behaviours affecting measurement quality through signal analysis, Python tooling and reproducible validation workflows.

- **Trace/result analysis:** Analysed measurements, simulations and diagnostic signals to identify abnormal behaviours and dominant contributors.
- **Debugging mindset:** Investigated environmental and instrumental effects impacting stability, sensitivity and performance.

PROFILE FOR CASTOR

Validation engineer with strong experience in system/software validation, anomaly analysis, logs/results interpretation, Python automation, Linux environments and hardware/software interface follow-up. Background developed in industrial connected infrastructure, operational software, space systems and complex technical validation environments.

For a **Connectivity Validation Engineer Freelance**

mission, I bring a strong validation mindset: test

plans, validation campaigns, manual and automated checks, analysis of logs and defects, technical reporting, quality follow-up and collaboration with software, hardware and system stakeholders.

- Experience in validation, verification, integration and analysis of complex software/system workflows.
- Industrial exposure to connected systems, operational infrastructure and hardware/software interfaces.
- Strong Python, scripting, Linux, Git/GitLab and automation background for test tooling and repeatable analysis.
- Strong capacity for logs, traces, defects, anomalies, technical risks and root-cause-oriented investigation.
- Comfortable with validation plans, test strategy, campaign execution, reporting quality and improvement of test environments.
- International technical experience and professional English for documentation, collaboration and reporting.
- Ready to ramp up on wireless protocol validation tools for Wi-Fi, Bluetooth, GNSS, Cellular and NFC environments.

TARGET MISSION FIT

- Validation of connected-system functionalities and operational behaviours.
- Preparation of test plans, validation strategies and campaign execution.
- Manual and automated testing using Python/scripts and reproducible environments.
- Analysis of logs, traces, defects, abnormal behaviours and technical findings.
- Collaboration with firmware, hardware, software and system stakeholders.
- Improvement of test environments, validation workflows and documentation.
- Technical reporting, quality follow-up and issue prioritisation.

LANGUAGES

English



Spanish



EDUCATION

PhD in Physics

Université Côte d'Azur

📅 2018 – 2021

📍 Nice, France

Selective Graduate Program in Fundamental Physics (Magistère)

Université Paris-Saclay

📅 2016 – 2018

📍 Orsay, France

MSc in Astronomy and Astrophysics

- **Python tooling:** Developed Python pipelines for noise source identification, uncertainty estimation, modelling and automated analysis.
- **Validation reporting:** Produced reproducible technical analyses, structured reports and review material for international stakeholders.
- **Technical communication:** Presented complex findings clearly to multidisciplinary teams.

Junior R&D Engineer – Simulation, Diagnostics & Validation Workflows

ARTEMIS, Observatoire de la Côte d'Azur

📅 2018 – 2021

📍 Nice, France

Context: Software and analysis activities for gravitational-wave mission preparation, involving simulation, diagnostic analysis and validation of complex signal environments.

Objective: Develop robust analysis and validation workflows for demanding technical environments.

- **Simulation & validation:** Built simulation approaches for complex signal environments, uncertainty studies and large-scale datasets.
- **Consistency checks:** Compared simulated outputs, model behaviours and analysis results to assess consistency and robustness.
- **Diagnostic analysis:** Investigated noise sources through cross-sensor correlation analyses and system-level diagnostics.
- **Performance analysis:** Supported identification of limiting factors affecting mission-level performance and system sensitivity.
- **Scientific computing:** Developed Python-based analysis, modelling, documentation and reproducible workflows.

Observatoire de Paris / Université Paris-Saclay

📅 2017 – 2018

📍 Paris, France

BSc in Fundamental Physics

Université Paris-Saclay

📅 2015 – 2016

📍 Orsay, France

PROFESSIONAL TRAINING

Machine Learning in Python with scikit-learn

FUN MOOC – Inria

📅 2026

📍 France Université Numérique

Predictive modelling, supervised learning, model evaluation, cross-validation, metrics, pipelines and practical machine learning workflows with Python and scikit-learn.

Supervised Machine Learning (Regression & Classification)

DeepLearning.AI – Andrew Ng

📅 2020

📍 Coursera

Recherche reproductible : principes méthodologiques pour une science transparente

FUN MOOC – Inria

📅 2025

📍 France Université Numérique

Continuous Professional Training – Software, Data, AI and Systems

OpenClassrooms

📅 2024 – 2026

Python, SQL, Machine Learning, AI, Linux, JavaScript, TypeScript, Angular, Node.js, Django, REST APIs, C/C++, web development, algorithms and software engineering fundamentals.

PROJECTS

VEV – Connected EV Charging Infrastructure, Operational Data Flows & Validation

Industrial Software / Connected Hardware Infrastructure

📅 2025 – 2026

📍 Valbonne, France

Contribution to software integration, validation checks and operational follow-up for a CPMS platform connected to EV charging infrastructure.

Connectivity validation relevance: analysis of communication issues, operational data exchanges, service behaviour, anomalies, incident reports and hardware/software interface constraints.

Technical relevance: production follow-up, FTP data flows, operational monitoring, software integration, defect analysis, technical reporting and stakeholder coordination.

CNES / LISA – Validation Campaigns, Automation & Technical Reporting

ESA/NASA Space Mission – Large-scale International Collaboration

📅 2023 – 2025

📍 Nice, France

Coordination and structuring of software, data-processing and validation workflows for the LISA space mission within a large international engineering and scientific collaboration.

Validation relevance: test strategy logic, validation campaigns, consistency checks, requirements-to-results alignment, technical evidence, anomaly investigation, reporting and documentation.

Software relevance: Python platform development, GitLab workflows, reproducible environments, model integration, data-processing chains and versioned deliverables.

STRENGTHS

Validation connectivité & tests

- Connectivity Validation
- System Validation
- Software Validation
- Validation Strategy
- Test Plans
- Test Campaigns
- Manual Testing
- Automated Testing
- Test Execution
- Test Reports
- Defect Tracking
- Quality Follow-Up
- Continuous Improvement

Logs, debugging & anomalies

- Log Analysis
- Trace Analysis
- Anomaly Identification
- Defect Investigation
- Root-Cause Reasoning
- System Debugging
- Operational Diagnostics
- Technical Reporting
- Risk Identification
- Corrective Actions
- Performance Analysis

Connectivity & embedded environments

- Wi-Fi Awareness
- Bluetooth Awareness
- GNSS Awareness
- Cellular Awareness
- NFC Awareness
- Wireless Protocols Ramp-Up
- Embedded Systems Awareness
- Linux
- Android Embedded Awareness
- Hardware/Software Interfaces
- Firmware Team Interface
- Hardware Team Interface
- Software Team Interface
- Connected Infrastructure

Automation, software & tools

- Python
- Bash
- Scripting
- Test Automation
- Linux Environments
- Git / GitLab
- GitHub
- CI/CD
- Docker
- SQL
- C
- C++
- TypeScript
- Node.js
- Angular
- REST APIs

Collaboration & domains

- Technical English
- International Environment
- Stakeholder Coordination
- Cross-Functional Collaboration
- Agile
- Kanban
- Backlog Follow-Up
- Jira
- Confluence
- Zenhub
- Industrial Software
- EV Charging Infrastructure
- Operational Monitoring
- Production Follow-Up
- Critical Systems